

ICT600 CORRECTED STUDENT SCHEME

July 2026

Question-ordered model answers and marking guide

Scope	July 2026 final examination
Arrangement	Answers follow the original paper from Question 1 through Part B. Coding, diagrams, and written responses are integrated once at their correct positions.
Assessment	100 marks: Part A 70, Part B 30
Status	Corrected student reference; equivalent technically correct answers may receive credit.

How to use this scheme

Question-order rule

Attempt the paper first. Then check the matching answer in this document. Each answer appears once, in the same order as the question paper.

- Equivalent correct wording, diagrams, algorithms, and code may receive full credit when they meet the stated requirements.
- For coding, award method marks when a minor syntax error does not invalidate an otherwise correct solution.
- For design or illustration questions, the labelled diagram is the model answer; prose alone is not treated as a complete design.

Part A

Question 1(a)

8 marks

Define URL, HTTP, HTTPS, and Web Application.

Mark allocation

Awardable element	Marks
URL	2
HTTP	2
HTTPS	2
Web application	2
Total	8

Complete model answer

- URL: the address used to identify and locate a web resource.
- HTTP: the request-response protocol used between web clients and servers.
- HTTPS: HTTP carried over TLS to protect data in transit and authenticate the server.
- Web application: software accessed through a browser and delivered over a network.

Question 1(b)

10 marks

Identify FIVE categories of web applications.

Mark allocation

Awardable element	Marks
First category and explanation	2
Second	2
Third	2
Fourth	2
Fifth	2
Total	10

Complete model answer

- Any five with brief explanations: static, dynamic, e-commerce, portal, animated/interactive, or CMS-based web application.

Question 2(a)**6 marks****Define JavaScript, HTML Element, and HTML Attribute.****Mark allocation**

Awardable element	Marks
JavaScript	2
HTML element	2
HTML attribute	2
Total	6

Complete model answer

- JavaScript adds dynamic behaviour and interactivity.
- An HTML element is a complete component represented by tags and content, or a void element.
- An HTML attribute supplies additional information or configuration in a start tag.

Question 2(b)**6 marks****State THREE HTML formatting elements.****Mark allocation**

Awardable element	Marks
First tag and effect	2
Second	2
Third	2
Total	6

Complete model answer

- Any three correctly explained elements such as , , <i>, , <mark>, <small>, , <ins>, <sub>, or <sup>.

Question 3(a) - CREATE TABLE**8 marks****Create library_books with the required fields and automatic primary key.****Mark allocation**

Awardable element	Marks
CREATE TABLE and name	2
book_id AUTO_INCREMENT PRIMARY KEY	2
title	1
author	1
year_published	1
Parentheses/semicolon and valid syntax	1
Total	8

Complete model answer

```
CREATE TABLE library_books (  
    book_id INT AUTO_INCREMENT PRIMARY KEY,
```

```

title VARCHAR(255) NOT NULL,
author VARCHAR(255) NOT NULL,
year_published INT
);

```

Question 3(b) - five SELECT queries

10 marks

Write every required products query rather than only describing the approach.

Mark allocation

Awardable element	Marks
All rows	2
product_name and price	2
price greater than 50	2
Electronics category	2
Descending price order	2
Total	10

Complete model answer

```

SELECT * FROM products;

SELECT product_name, price FROM products;

SELECT * FROM products WHERE price > 50;

SELECT * FROM products WHERE category = 'Electronics';

SELECT * FROM products ORDER BY price DESC;

```

Question 4(a)

4 marks

Identify FOUR popular web application frameworks.

Mark allocation

Awardable element	Marks
First framework	1
Second	1
Third	1
Fourth	1
Total	4

Complete model answer

- Any four valid frameworks, for example Laravel, CodeIgniter, CakePHP, Django, Ruby on Rails, Spring, ASP.NET Core MVC, or Express.

Question 4(b)**6 marks****Explain THREE challenges of using a framework for a small corporate-profile project.****Mark allocation**

Awardable element	Marks
First challenge and project relevance	2
Second	2
Third	2
Total	6

Complete model answer

- Learning curve may be disproportionate to the small project.
- Framework overhead may add unnecessary complexity and resource use.
- Dependencies and version/security maintenance create ongoing work.
- Reduced flexibility is another valid challenge when accurately explained.

Question 5(a)**6 marks****Classify THREE common web vulnerabilities relevant to malicious login input.****Mark allocation**

Awardable element	Marks
First vulnerability and scenario link	2
Second	2
Third	2
Total	6

Complete model answer

- SQL injection
- Cross-site scripting (XSS)
- Broken authentication or session management

Question 5(b)**6 marks****Assess THREE malware types relevant to a suspicious attachment and unwanted advertisements.****Mark allocation**

Awardable element	Marks
First malware type and assessment	2
Second	2
Third	2
Total	6

Complete model answer

- Trojan horse: a disguised malicious attachment. Adware: software that produces unwanted advertisements. Spyware: software that covertly collects information.

Part B

Part B Question 1 - BMI calculator

15 marks

Provide the complete runnable HTML/JavaScript BMI calculator.

Mark allocation

Awardable element	Marks
HTML document	2
Inputs, button and result	3
Function/event	2
Empty/numeric/positive validation	3
BMI formula	2
Category logic	2
Displayed result	1
Total	15

Complete model answer

```

<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>BMI Calculator</title>
</head>
<body>
  <h1>BMI Calculator</h1>
  <label>Weight (kg): <input id="weight" type="number" min="1" step="0.1"></label>
  <label>Height (m): <input id="height" type="number" min="0.1" step="0.01"></label>
  <button type="button" onclick="calculateBMI()">Calculate</button>
  <p id="result" aria-live="polite"></p>

  <script>
    function calculateBMI() {
      const weight = Number(document.getElementById('weight').value);
      const height = Number(document.getElementById('height').value);
      const result = document.getElementById('result');

      if (!Number.isFinite(weight) || !Number.isFinite(height) ||
        weight <= 0 || height <= 0) {
        result.textContent = 'Enter positive numeric values.';
        return;
      }

      const bmi = weight / (height * height);
      let category;
      if (bmi < 18.5) category = 'Underweight';
      else if (bmi < 25) category = 'Normal';
      else if (bmi < 30) category = 'Overweight';
      else category = 'Obese';

      result.textContent = `BMI: ${bmi.toFixed(2)} (${category})`;
    }
  </script>
</body>
</html>

```

Part B Question 2 - Student Registration Form

15 marks

Provide the complete HTML/PHP form, validation and result display.

Mark allocation

Awardable element	Marks
Form structure	3
Name, ID and programme	3
Gender controls	2
POST handling	2
Empty-value validation	3
Success/error output	2
Total	15

Complete model answer

```
<?php
$name = $studentId = $gender = $programme = '';
$message = '';
$submitted = false;

function e(string $value): string {
    return htmlspecialchars($value, ENT_QUOTES, 'UTF-8');
}

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $name = trim($_POST['name'] ?? '');
    $studentId = trim($_POST['student_id'] ?? '');
    $gender = trim($_POST['gender'] ?? '');
    $programme = trim($_POST['programme'] ?? '');

    if (empty($name) || empty($studentId) ||
        empty($gender) || empty($programme)) {
        $message = 'Unsuccessful: all fields are required.';
    } else {
        $submitted = true;
        $message = 'Registration successful.';
    }
}
?>
<!doctype html>
<html lang="en">
<head>
    <meta charset="utf-8">
    <title>Student Registration</title>
</head>
<body>
    <h1>Student Registration Form</h1>

    <form method="post" action="">
        <label>Name: <input name="name" value="<?= e($name) ?>" required></label><br>
        <label>Student ID: <input name="student_id" value="<?= e($studentId) ?>" required></label><br>
        <span>Gender:</span>
        <label><input type="radio" name="gender" value="Male" required
            <?= $gender === 'Male' ? 'checked' : '' ?>> Male</label>
        <label><input type="radio" name="gender" value="Female"
            <?= $gender === 'Female' ? 'checked' : '' ?>> Female</label><br>
        <label>Programme: <input name="programme" value="<?= e($programme) ?>" required></label><br>
        <button type="submit">Register</button>
    </form>

    <?php if ($message !== ''): ?>
        <p><?= e($message) ?></p>
    <?php endif; ?>

    <?php if ($submitted): ?>
        <h2>Submitted Details</h2>
        <p>Name: <?= e($name) ?></p>
        <p>Student ID: <?= e($studentId) ?></p>
        <p>Gender: <?= e($gender) ?></p>
        <p>Programme: <?= e($programme) ?></p>
    <?php endif; ?>
</body>
</html>
```